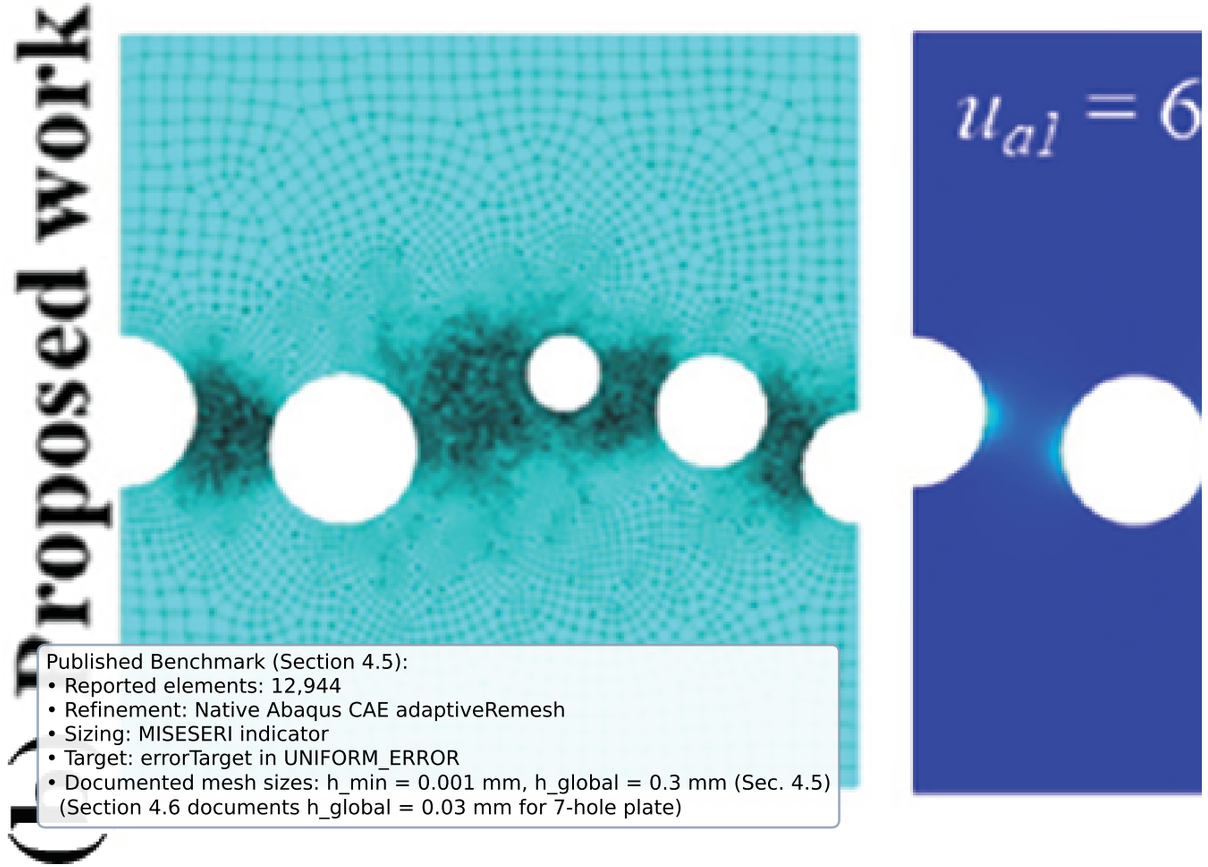
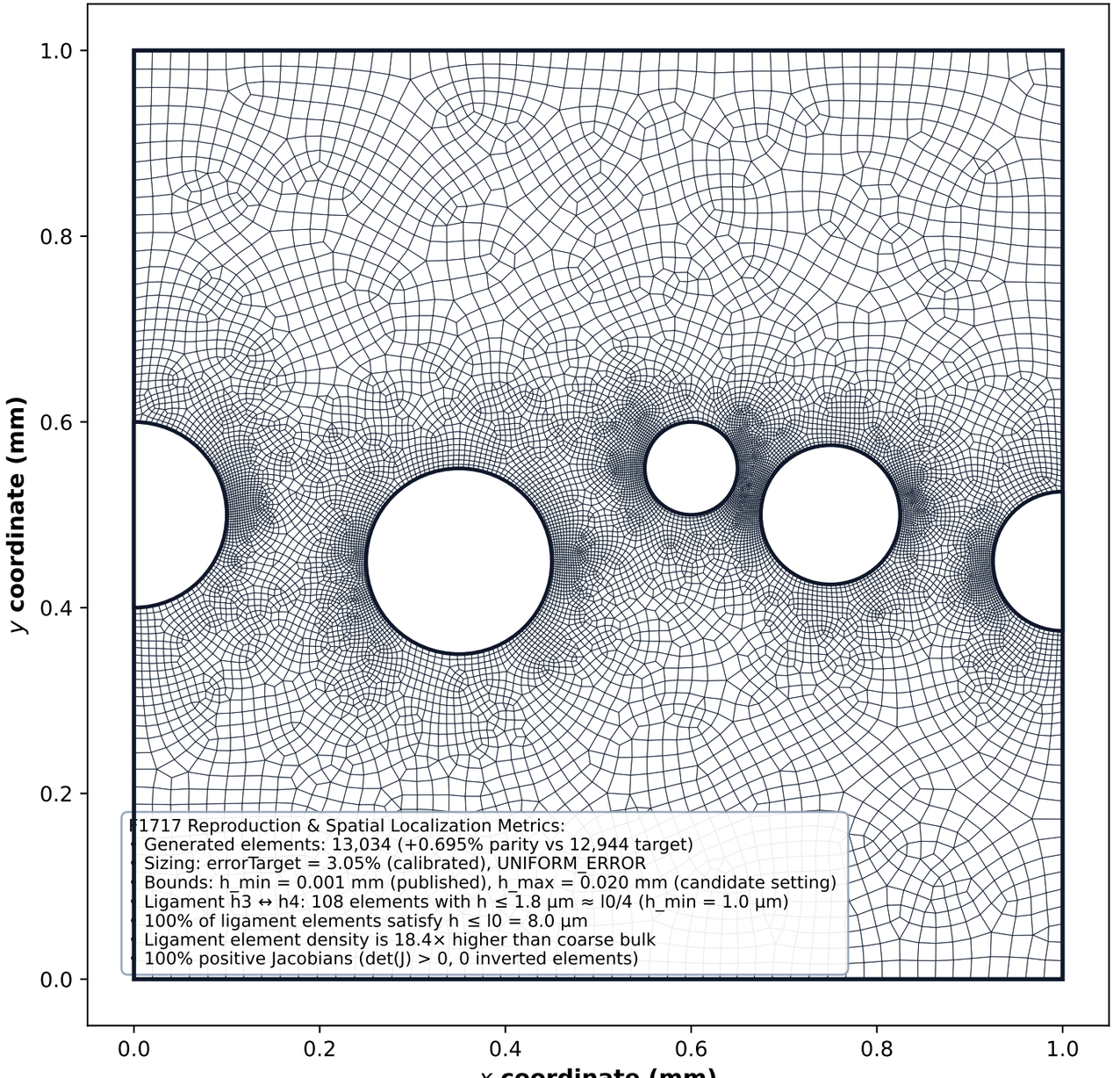
(a) Published Proposed PFM Adaptive Mesh
(Pandey & Kumar 2025, Fig. 23b, 12,944 Elements)



Published Benchmark (Section 4.5):
- Reported elements: 12,944
- Refinement: Native Abaqus CAE adaptiveRemesh
- Sizing: MISESERI indicator
- Target: errorTarget in UNIFORM_ERROR
- Documented mesh sizes: h_min = 0.001 mm, h_global = 0.3 mm (Sec. 4.5)
  (Section 4.6 documents h_global = 0.03 mm for 7-hole plate)

(b) Reproducing Native Adapted Mesh (Task #1717)
(13,034 Elements, +0.70% Parity vs Published 12,944)



#1717 Reproduction & Spatial Localization Metrics:
Generated elements: 13,034 (+0.695% parity vs 12,944 target)
Sizing: errorTarget = 3.05% (calibrated), UNIFORM_ERROR
Bounds: h_min = 0.001 mm (published), h_max = 0.020 mm (candidate setting)
Ligament h3 ↔ h4: 108 elements with h ≤ 1.8 μm ≈ l0/4 (h_min = 1.0 μm)
100% of ligament elements satisfy h ≤ l0 = 8.0 μm
Ligament element density is 18.4× higher than coarse bulk
100% positive Jacobians (det(J) > 0, 0 inverted elements)